

# Vibhum Sharma

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[linkedin.com/in/vibhum-sharma-10j](https://www.linkedin.com/in/vibhum-sharma-10j) | [github.com/rovermask](https://github.com/rovermask) | [vibhumsharma.vercel.app](https://vibhumsharma.vercel.app)

## SUMMARY

Ph.D. research scholar in Information Technology at IIIT Allahabad working on multimodal disease prognosis, with hands-on experience in machine learning, deep learning (CNN), computer vision, NLP and Python backend development (Flask, Django, REST APIs). Built and deployed end-to-end ML applications, including a tuberculosis detection model with 97.7% accuracy (patent application filed), and completed internships at ParallelDots and SmartED. Seeking Machine Learning Engineer, AI Engineer, Data Scientist, Data Analyst and Python Backend Developer roles.

## EDUCATION

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|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2026 – Present   | Ph.D. in Information Technology (Full-time), <b>Indian Institute of Information Technology Allahabad (IIIT Allahabad)</b> , Prayagraj<br><i>Research area: Multimodal Disease Prognosis</i> |
| Sept 2023 – May 2025 | Master of Computer Applications (MCA), <b>Sam Higginbottom University of Agriculture, Technology and Sciences (SHUATS)</b> , Prayagraj<br><i>CGPA: 7.88/10</i>                              |
| Jul 2020 – Jul 2023  | Bachelor of Computer Applications (BCA), <b>Ewing Christian College</b> , Prayagraj                                                                                                         |

## TECHNICAL SKILLS

|                                           |                                                                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>                         | Python, SQL, C++, JavaScript                                                                                                                                                                                                        |
| <b>Machine Learning &amp; AI:</b>         | Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), Computer Vision, Natural Language Processing (NLP), TensorFlow, Keras, scikit-learn, OpenCV, MediaPipe, Data Augmentation, Model Evaluation, Model Deployment |
| <b>Data Analysis &amp; Visualization:</b> | pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Cleaning, Feature Extraction                                                                                                                              |
| <b>Backend &amp; Web:</b>                 | Flask, Django, REST APIs, API Testing, React, HTML, CSS                                                                                                                                                                             |
| <b>Databases:</b>                         | MySQL, Firebase (Firestore)                                                                                                                                                                                                         |
| <b>Cloud &amp; Tools:</b>                 | Microsoft Azure, Vercel, Render, Git, GitHub                                                                                                                                                                                        |

## WORK EXPERIENCE

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|---------------------------------------------|---------------------|
| <b>Backend Intern, ParallelDots, Remote</b> | Oct 2025 – Jan 2026 |
|---------------------------------------------|---------------------|

- Designed and executed automated and manual test cases for RESTful backend APIs powering the ShelfWatch product, improving API reliability and regression coverage.
- Collaborated with backend engineers to validate edge cases, error handling and performance across production-critical endpoints.
- Worked with Python in a production backend environment and followed quality assurance practices for API testing.

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|------------------------------------------------------------------|----------------------|
| <b>Data Science and Machine Learning Intern, SmartED, Remote</b> | Sept 2024 – Dec 2024 |
|------------------------------------------------------------------|----------------------|

- Built an end-to-end sentiment analysis pipeline using NLP techniques, covering data cleaning, feature extraction, model training and evaluation.
- Performed exploratory data analysis (EDA) and visualization with pandas and Seaborn to derive insights from social media datasets.
- Applied text preprocessing techniques including tokenization, stemming and vectorization for machine learning models.

# PROJECTS

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**LungVision: Tuberculosis Detection from Chest X-rays** Feb 2025 – Apr 2025  
*TensorFlow, Keras, CNN, Computer Vision, Flask, Render* | [GitHub](#) | [Live Demo](#)

- Designed, trained and evaluated a Convolutional Neural Network (CNN) to classify chest X-ray images as tuberculosis-positive or normal, achieving 97.7% accuracy using preprocessing and data augmentation.
- Deployed the model as a Flask web service on Render for real-time inference; visualized accuracy curves and confusion matrix with Matplotlib and Seaborn.
- Filed a patent application for the CNN-based tuberculosis detection system.

**WatchBuddy: Movie, Series and Book Tracker** Jul 2025 – Sept 2025  
*React, Firebase Authentication, Firestore, REST APIs* | [GitHub](#) | [Live Demo](#)

- Built a full-stack web application with React and Firebase (Authentication, Firestore) to track movies, series and books, integrating the TMDB and Google Books REST APIs with live search suggestions.
- Implemented ISBN barcode scanning, custom lists, personalized recommendations and a light/dark theme; deployed on Vercel.

**AirCanvas: Gesture-Controlled Virtual Drawing** Sept 2024 – Dec 2024  
*Python, OpenCV, MediaPipe* | [GitHub](#)

- Implemented a real-time virtual drawing application using hand-gesture recognition from webcam input, and optimized gesture tracking and rendering for smooth, accurate strokes.

# CERTIFICATIONS

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**Microsoft Certified: Azure AI Fundamentals (AI-900)**, Microsoft Sept 2025  
**Agentic AI Development**, Eduxlabs, IIT Madras Aug 2025  
**Python for Data Science and Machine Learning Bootcamp**, Udemy Mar 2025  
**Data Analytics and Visualization Job Simulation**, Accenture (Forage) Jan 2025  
**Machine Learning Internship Certificate**, SmartED Dec 2024  
**Introduction to Python**, Microsoft Technology Associate (MTA) Aug 2021  
**HackerRank Certificates:** Python (Basic), SQL (Basic), Problem Solving (Basic and Intermediate)